

Aditya Kulkarni

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github.com/Aditya-k24 | Raleigh, NC (Remote)

PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with production cloud platform engineering experience across REST microservices, distributed systems, and cloud-native deployments. At Isomer AI, owned end-to-end delivery of 10+ backend features across TypeScript/Node.js/PostgreSQL/Redis -- 394 commits, 7 production deployments with Docker, CI/CD, and Sentry observability. Built Java 21 + Spring Boot transactional services with Testcontainers concurrency testing, and event-driven orchestration with Temporal.io and Kafka at 2,000 async requests/second. Proficient in Java, Python, TypeScript, and AWS; experienced in object-oriented design, agile workflows, code reviews, and production on-call operations.

CORE COMPETENCIES

Cloud-Native Microservices

Java / Python / TypeScript

REST API Design

Distributed Systems

AWS / GCP Deployments

Object-Oriented Design

CI/CD Pipelines

Agile Development

WORK EXPERIENCE

Isomer AI

May 2025 - Present

Product Engineer Intern — Chicago, IL (Remote)

- Designed and shipped 10+ **cloud-native REST microservices** in TypeScript/Node.js/PostgreSQL/Redis -- RBAC, real-time event pipelines (Pusher), and Temporal.io workflow orchestration -- across 7 production deployments with Docker and CI/CD
- Built document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 storage) and multi-provider LLM agent tooling; contributed 394 commits and reviewed pull requests across the engineering team; monitored production with Sentry

Mphasis (HP Company)

Jun 2024 - Aug 2024

Software Developer Intern — Mumbai, India

- Automated deployments on **AWS (EC2, S3, CodeDeploy)** and **GCP**; integrated Grafana monitoring dashboards and supported model validation and red-teaming for production reliability
- Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML; improved claim accuracy by 25%

Capgemini

Jun 2023 - Aug 2023

AI/ML Intern — Mumbai, India

- Led end-to-end SDLC for a distributed data platform automating support ticket analysis, reducing processing time by 15%
- Implemented **Hadoop + Spark** batch pipelines and **Kafka** streaming for real-time ingestion, reducing ticket-processing latency by 20%

NC State University

Apr 2025 - Present

Graduate Research Assistant — Raleigh, NC

- Designed a cloud-based genomic data platform with JBrowse 2 and MERN stack, enabling exploration of 500GB datasets for 100+ researchers worldwide, deployed on **Google Cloud Run**
- Applied hyperparameter tuning and time-series analysis to ML architecture, improving model accuracy by 12%

PROJECTS

Inventory Reservation Service Java 21 + Spring Boot

Built a transactional inventory reservation API preventing overselling via PostgreSQL row-level locking (SELECT FOR UPDATE) across multi-SKU reservations; idempotent REST endpoints with RFC 9457 error responses and automated expiry via scheduled cleanup. Verified concurrency correctness with Testcontainers -- two concurrent requests against quantity 1 produce exactly one success and one 409. Flyway migrations, Spring Security (role-based), multi-stage Docker, GitHub Actions CI.

Java 21, Spring Boot 3.5, PostgreSQL, Spring Data JPA, Flyway, Spring Security, Testcontainers, Docker

AlgoPulse – Cloud AI Orchestration Platform 2025

Identified API-polling throughput bottleneck under load and re-architected to event-driven model -- Supabase Edge Functions stream AI reasoning loops directly to the client. Temporal.io and Kafka handle durable workflow state and fault-tolerant worker coordination: 2,000 async requests/second without state loss.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

LLM Eval Harness Python

Vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across 2,000-example frozen test set. Metrics: ExactMatch, TokenF1, grounded hallucination rate, p50/p95 latency, and cost/call with paired bootstrap CIs and McNemar testing. Stack: Python, DuckDB, Parquet, Streamlit, Docker.

Python, Anthropic/OpenAI/Google SDKs, DuckDB, Parquet, Streamlit, Docker

Demand Paging – Xinu OS Systems

Engineered an 8-store virtual memory system in C with Second-Chance page replacement; reduced page faults by 25%. Added 7+ kernel-level system calls; improved memory performance by 15% via inverted page tables and custom fault handler.

C, Xinu OS, Virtual Memory, Kernel Development

EDUCATION

North Carolina State University — MS Computer Science

Aug 2024 - May 2026

Courses: Artificial Intelligence, Algorithms, Data Science, Operating Systems, Software Engineering, Object-Oriented Development

Dwarkadas J. Sanghvi College of Engineering — BTech Computer Science

Aug 2020 - May 2024

CGPA: 9.32/10 · Courses: Data Warehousing, Big Data Infrastructure, DS & Algorithms, Machine Learning, Software Engineering

SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, C, R, Ruby

Frameworks & Tools: Spring Boot, Node.js, React, Temporal.io, Kafka, Spark, Hadoop, Docker, Kubernetes, Git, GitHub Actions

Cloud & Infra: AWS (EC2, S3, CodeDeploy, CloudFormation), GCP (Cloud Run), Grafana, CI/CD Pipelines, Distributed Systems

Databases: PostgreSQL, MySQL, MongoDB, Redis, NoSQL

AI/ML: OpenAI API, Anthropic Claude, LangChain, MCP, Hugging Face, TensorFlow, PyTorch, RAG, Pandas, NumPy